Simulating Climate Risk Migration:
The Shadow Banking Spillover Effect

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Contents

1. Abstract
2. Climate Risk in Banking
3. Simulating Climate Risk Migration
    a. Idea
    b. Methodology: Discussing the four phases
    c. Discussion
4. Bibliography

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1. Abstract

Climate change has been a topic of discussion for a few decades now, but it has seen a significant upward trend since the beginning of the 21st century. The banking and financial sector has not remained untouched by the effects of climate change. As the governments and institutions across the world implement and discuss policies such as Carbon Tax and Energy Transition, the financial sector is being forced to adapt to a rapidly shifting landscape of both physical and transition risks. However, while regulatory bodies are pushing standard commercial banks to strictly price these climate risks into their lending models, the systemic consequences of such strict regulations are often overlooked. This is a two-part project; the first part presents the scattered ideas and concepts of climate risk in banking in a consolidated manner in form of a review; the second part presents a macroeconomic perspective of strict climate regulations on traditional banks. I do so by a novel simulation based approach through which I show that, in the case of strict climate regulations, the case in which traditional banks correctly price their climate risk, the systemic climate risk is not eliminated - the risk is migrated to the shadow banking sector, including NBFCs. I then show that in case of a climate shock, inclusion of climate risk in banks' lending models deprives firms of much needed debt, and thus forces firms to borrow from the shadow banking sector, including NBFCs, at a much higher climate premium, leading to high default rates. Thus, showing that correct pricing of climate risk by banks does not eliminate the systemic climate risk from a macroeconomic perspective.

The project report is arranged as follows: Firstly, I discuss how climate change leads to risk in banking. Then, I move on to discuss the novel simulation based approach to show that the systematic climate risk is not mitigated from a macroeconomic perspective just by inclusion of such risks in banks' lending models. I discuss the idea, methodology and results of the simulation exercise.

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2. Climate Risk in Banking

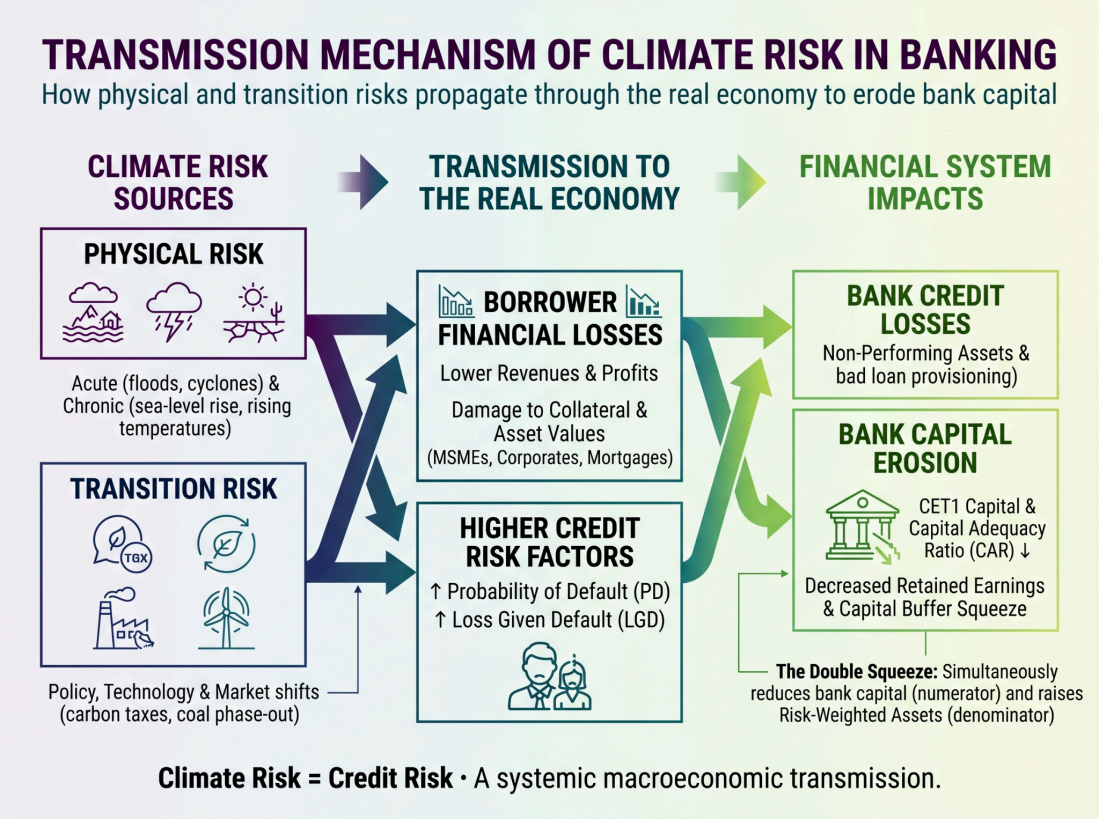


Figure 1: Transmission Mechanism of Climate Risk in Banking
(This is an AI-generated image made using Google Gemini)

Climate change means a long-term shift in temperature and weather patterns. For an economy, this shows up in two different ways. First, the weather itself becomes more extreme: more floods, stronger cyclones, longer heatwaves, and droughts that last longer. Second, governments and societies try to slow climate change down, and they do this through carbon taxes, rules that phase out fossil fuels, and cheaper clean technology. Both of these things change how much money firms and households make.

Climate risk is the chance that one of these two things hurts someone financially. **Financial climate risk** is what we get when that hurt travels into the financial system, that is, into banks, insurers and investors. The simplest way to think about it is this: a bank does not care about the weather as such. It cares about whether the people it has lent money to will keep paying it back.

2.1 Why should a Bank care?

A bank earns its money by lending. It gives loans to companies, to households buying homes, to small businesses and to farmers, and it expects to get the money back with interest. Think about what those borrowers are doing. A power company runs a coal plant. A family has a

mortgage on a house in a flood-prone area. A small manufacturer has a unit near a river. A farmer depends on the monsoon. The bank does not own any of these assets, but its money is tied up in all of them.

Crucially, banks do not generally face climate risk *directly* from weather events. A bank's headquarters being flooded is an operational risk, but it is not the main problem. The true danger is indirect: banks are exposed to climate risk through the real economy. They lend to corporate sectors (like power, steel, cement, and transport), retail mortgages in flood- or heat-prone regions, and Micro, Small, and Medium Enterprises (MSMEs) whose livelihoods depend on stable weather patterns (like agriculture). Therefore, when a climate shock hits a borrower, it deteriorates that borrower's financial health, which in turn hits the bank. In short: **Climate Risk equals Credit Risk.**

The main places where a typical bank is exposed are:

- Corporate loans, especially to power, steel, cement and transport.
- Retail mortgages, especially in flood-prone or heat-prone regions.
- MSME and agricultural lending, where income depends directly on the weather.
- The bank's own investment portfolio of bonds and equities.

2.2 Climate Risk: Physical Risk and Transition Risk

Almost all of the literature divides climate-related financial risk into two groups. The difference between them matters because they hit different borrowers at different times.

Physical risk

Physical risk is the loss that comes from the climate itself. It has two kinds. **Acute** risks are sudden events: a flood damaging houses and small factories, a cyclone hitting coastal ports, a heatwave cutting crop output. **Chronic** risks are slow changes that build up over years: rising sea levels pulling down real estate values near the coast, or persistent drought weakening the income of rural borrowers. For a bank, the damage usually comes through the borrower's income and through the value of the collateral, since a damaged house or factory is worth less if the bank has to sell it.

Transition risk

Transition risk is the loss that comes from the shift to a low-carbon economy. Nothing physically breaks, but the rules of the game change. Carbon taxes raise operating costs, fossil fuel phase-out rules shut down certain business lines, and technology moves on (for example, renewables replacing coal). A firm that was profitable last year can quickly become unprofitable. Some assets lose their value permanently, and these are called **stranded assets**. A coal mine under a phase-out rule, a thermal power plant displaced by solar, and oil reserves that are no longer worth extracting are all common examples.

	Physical Risk	Transition Risk
Where it comes from?	Extreme weather and long-term climate change	Policy, technology and market shifts towards low carbon
Examples	Flood, cyclones, heatwaves, sea-level rise, drought	Carbon tax, coal phase-out, renewables replacing coal
Who is hit most?	Borrowers in vulnerable locations (coasts, flood plains, farms)	Borrowers in carbon-intensive sector (power, steel, cement)
How it reaches the bank?	Lower income and damaged collateral, so higher PD and LGD	Lower profits and stranded assets, so higher PD and LGD

Table 1: Comparison between Physical and Transition risk

(PD: Probability of default, LGD: Loss given default)

Here is a simple example of transition risk. Suppose a bank has financed a coal power plant worth Rs 5,000 crore. A new carbon tax reduces the plant's EBITDA (its operating profit) by 30%. If the plant had only a small cushion between its profit and its debt payments, this can push its debt service coverage ratio below 1, which means the plant no longer earns enough to pay what it owes. The chance of default has gone up. The coverage idea used in this example is exactly the one I use in the simulation in Section 3, where I call it the Interest Coverage Ratio.

2.3 Measurement of Climate Risk

To measure how much a climate event can hurt a bank, we need three simple ideas from credit risk. Consider lending Rs 100:

- **Probability of Default (PD):** the chance that the borrower will not repay. If there is a 3 in 100 chance that the borrower will not pay back, PD is 3%.
- **Loss Given Default (LGD):** if the borrower defaults, what share of the money is actually lost, after whatever can be recovered (for example, by selling collateral). If Rs 40 can be recovered of the Rs 100, then LGD is 60%.
- **Exposure at Default (EAD):** the amount that is still outstanding when the default happens. Here, Rs 100.

Putting the three together gives the **Expected Loss (EL)**, which is the average loss the bank should expect on a loan or a portfolio:

$$EL = PD \times LGD \times EAD$$

A climate shock makes this number go up in two ways. It raises PD because the borrower is now weaker, and it can raise LGD because the collateral is now worth less. The chain looks like this:

$$Climate\ shock \rightarrow Firm\ losses \rightarrow Higher\ PD/LGD \rightarrow Bank\ credit\ losses$$

Consider the following examples:

Example	Exposure	PD (before to after)	LGD (before to after)	EL (before/after)	Increase in EL
Flood destroys a manufacturing unit	500	3% to 15%	60% (both)	9 / 45	36
Mortgages in a flood-prone region	15,000	2% to 5%	30% to 50%	90 / 375	285
Carbon policy hits a carbon-intensive sector	20,000	4% to 8%	50% (both)	400 / 800	400

Table 2: The first and the third rows only change PD, while the second row changes both PD and LGD, which is why the loss jumps by more than four times, even though PD only rises from 2% to 5%. In the second example, the flood does two things at once: borrowers find it harder to pay, and the houses that back the loans become worth less.

2.4 What this does to a bank's balance sheet?

An increase in expected loss is not just a number on paper. It travels through the bank's accounts in a fixed order. As more borrowers stop paying, the bank's **non-performing assets (NPAs)** rise. Rules require the bank to set aside money against these bad loans, which is called **provisioning**, and provisions are charged against profit. Lower profit means less retained earnings, and so the bank's capital gets smaller.

Capital is important because it is the cushion that absorbs losses before depositors are affected. Three terms are used all the time:

- **CET1 (Common Equity Tier 1):** the bank's highest-quality capital, mostly shareholders' equity and retained earnings. It is the first line of defence.
- **RWA (Risk-Weighted Assets):** the bank's assets, each counted according to how risky it is. A riskier loan carries a higher weight, so it counts for more.
- **CAR (Capital Adequacy Ratio):** capital divided by RWA. Regulators set a minimum, and the bank must always stay above it.

$$Climate\ shock \rightarrow Borrower\ loss \rightarrow Higher\ expected\ loss \rightarrow Capital\ erosion$$

Consider the transition example from *Table 2*. The bank has Rs 20,000 crore lent to a carbon-intensive sector, and the extra expected loss is Rs 400 crore. If the bank's CET1 capital is Rs 10,000 crore, then it falls to Rs 9,600 crore. In another simple case, a bank with CET1 of Rs 20,000 crore and RWA of Rs 1,50,000 crore has a CET1 ratio of 13.3%. If climate losses of Rs 2,000 crore build up, CET1 falls to Rs 18,000 crore and the ratio drops to 12%. Even a moderate climate shock reduces the buffer.

The double squeeze on capital

Climate risk can also hit the capital ratio from the other side, through the denominator. When a carbon-intensive borrower is downgraded, its risk weight goes up, and so does the bank's RWA. Say a Rs 5,000 crore exposure moves from a 100% to a 120% risk weight. RWA goes from Rs 5,000 crore to Rs 6,000 crore. With a required CAR of 11.5%, the capital needed against this exposure rises from Rs 575 crore to Rs 690 crore, so the bank needs Rs 115 crore of extra capital without having lost a single rupee yet.

Now put both effects together. If the same borrower's higher default probability also causes a Rs 100 crore loss, capital falls from Rs 575 crore to Rs 475 crore, while RWA rises to Rs 6,000 crore. The ratio becomes 475 / 6,000, which is about 7.9%, below the 11.5% requirement. This is why climate risk is called a prudential capital issue, and not only a sustainability reporting issue: it pushes the numerator down and the denominator up at the same time.

2.5 Why it is a system wide problem?

One bank taking one climate loss is manageable. The bigger worry is what happens when many banks are hit in the same way. There are three reasons why this can happen.

- **Concentration.** Banks tend to lend heavily to the same sectors (power, infrastructure, steel, cement), so a single policy change can hurt every bank at once.
- **Amplification.** Losses can add up across sectors. For instance, a bank with CET1 of Rs 18,000 crore and Rs 13,000 crore of stressed exposure in steel and cement (average PD increase of 4%, LGD of 50%) loses another Rs 260 crore, and CET1 falls to Rs 17,740 crore. Each shock looks small. Repeated shocks across sectors are not.
- **Credit tightening.** When banks are under pressure, they protect themselves by charging more and by lending less. This hurts the real economy, because firms that cannot get credit are weaker, which creates more losses. Climate stress can therefore be procyclical.

The last point is the one I care about most for this report. If a bank cuts credit to weaker borrowers to protect its own balance sheet, those borrowers do not disappear. They still need money, and they will look for it elsewhere. Section 3 follows exactly that question.

2.6 Single and Double Materiality

Regulators decide what a bank must report by asking whether something is **material**, meaning important enough to change the decisions of investors, regulators or other stakeholders. Traditionally, the test was **single (financial) materiality**: does this factor affect the firm's own earnings, capital or solvency? Climate risk entered banking regulation once it became financially material for banks.

The newer approach is **double materiality**, which asks a second question as well. Besides "how does climate affect the bank?", it asks "how does the bank's lending affect the climate?". For example, if a bank lends Rs 20,000 crore to coal power projects and each project emits 1 million tons of CO2 a year, its lending is indirectly linked to 20 million tons of emissions. At the same time, a transition policy that raises the PD of those loans is a financial risk to the bank. The bank is both exposed to climate risk and a contributor to it. The EU's sustainable finance rules, including the ***Corporate Sustainability Reporting Directive (CSRD)***, are built on this idea.

2.7 How regulators and banks are responding?

A lot of work is going into building tools that measure and disclose climate risk. Here, I present an overview of some of the prominent policies and initiatives.

Scenarios and stress testing

The **Network for Greening the Financial System (NGFS)** is a voluntary group of central banks and supervisors, set up in 2017. As of January 2026 it had 149 members and 24 observers. It does not make binding rules, but it publishes common climate scenarios that national regulators adapt. There are three broad kinds:

Scenario	What it looks like?
Orderly transition	Early policy action and gradual carbon pricing, so the transition shock is small.
Disorderly transition	Action is delayed and then sudden, with a sharp carbon price spike and a rapid repricing of assets.
Hot house world	Little mitigation, so the physical damage in the long run is severe.

Table 3: NGFS climate scenarios

A **climate stress test** takes one of these scenarios and asks what it would do to a bank's PD, LGD, exposures and capital ratios. As an illustration, take a bank called "Bank India" with Rs 10,000 crore lent to thermal power. A sudden carbon tax raises the PD of those borrowers from 2% to 6% (LGD stays at 50%). Expected loss rises from Rs 100 crore to Rs 300 crore, an increase of Rs 200 crore. If CET1 was Rs 18,000 crore against RWA of Rs 1,50,000 crore,

the CET1 ratio moves from 12% to 11.87%. It is a mild decline, but if similar shocks hit several sectors, the effect compounds.

A related tool is **Climate Value at Risk (Climate VaR)**. The two are often confused, but are actually different: Climate VaR estimates *how much could be lost*, while a climate stress test evaluates *whether the bank can survive that loss*. A big challenge for both is the time mismatch, since climate scenarios run for 20 to 30 years while supervisory stress tests look at 1 to 3 years.

Disclosure

Markets and regulators cannot judge climate risk unless banks and firms report it. The **Task Force on Climate-related Financial Disclosures (TCFD)**, created by the **Financial Stability Board (FSB)**, set out four pillars: governance, strategy, risk management, and metrics and targets. It became the global baseline and was later built into the **International Sustainability Standards Board's (ISSB) International Financial Reporting Standards (IFRS) S2** climate standard, which also asks for Scope 1, 2 and 3 emissions and scenario analysis. The **EU's CSRD** makes such reporting legally binding.

The Indian context:

India has been moving in the same direction. **SEBI's Business Responsibility and Sustainability Reporting (BRSR)**, made mandatory from 2023 for the top 1,000 listed firms, requires disclosure of emissions, energy use and ESG risks. The **RBI's 2022 discussion paper on climate risk and sustainable finance** treats climate risk as a financial stability concern, and the RBI's later guidance (2023 onwards) asks regulated entities, which include scheduled commercial banks and large NBFCs, to bring climate risk into their governance, risk management and capital assessment **(ICAAP, Internal Capital Adequacy Assessment Process)**. This is in line with the Basel Committee's principles.

Measuring financed emissions and classifying green activity

For a bank, most of its carbon footprint does not come from its own offices (Scope 1 and 2) but from the firms it finances (Scope 3). This is called **financed emissions**. For example, if a bank has lent to a steel company that emits 5 million tons of CO2 a year, and the bank's share of that company's financing is 20%, its financed emissions are $0.20 \times 5 = 1$ million tons. A **climate taxonomy** is a rulebook that says which activities count as green (solar, wind), transition (for example, natural gas under conditions) or brown (coal mining). The EU makes banks report a **Green Asset Ratio (GAR)**, which is taxonomy-aligned assets divided by total eligible assets; a bank with Euro 100 billion of aligned assets out of Euro 500 billion has a GAR of 20%.

Now, I will move on to the second half of my project which is "Simulating Climate Risk Migration"

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3. Simulating Climate Risk Migration

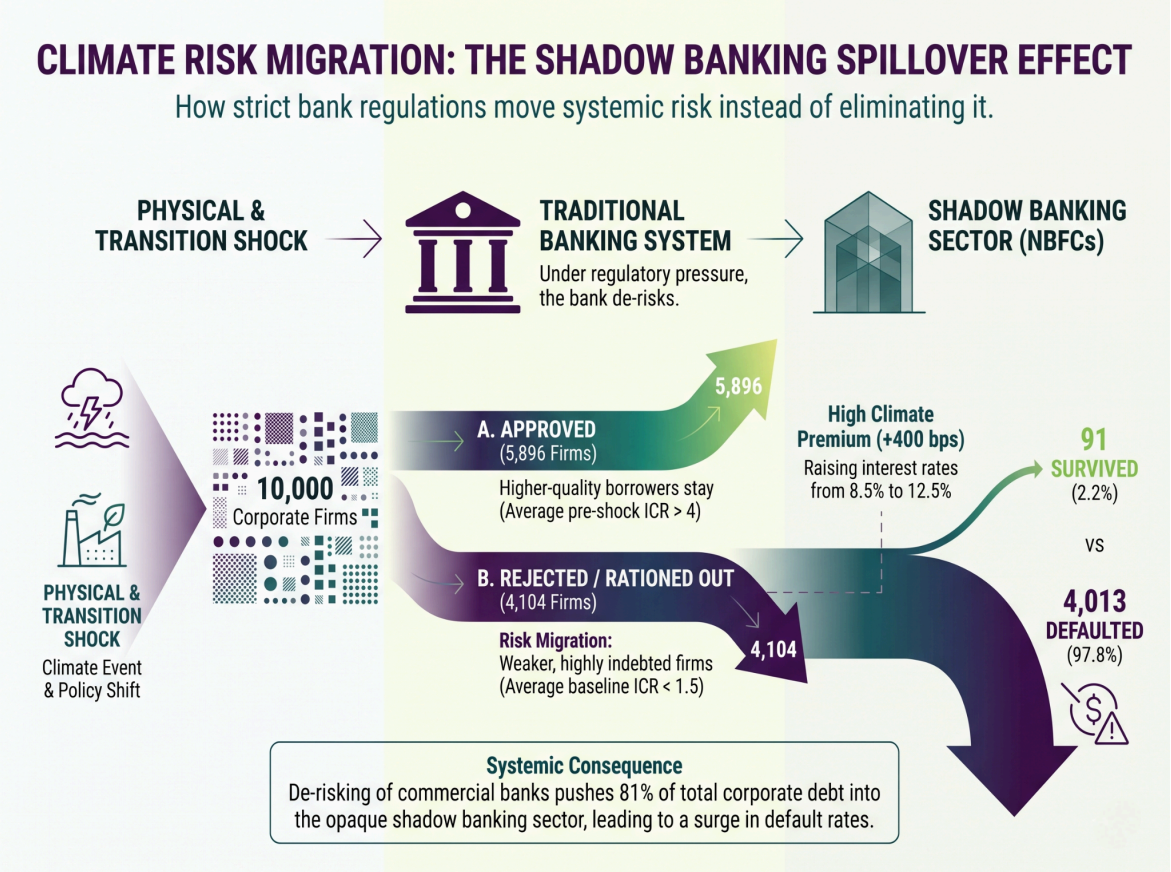


Figure 2: Climate Risk Migration

(This is an AI-generated image made using Google Gemini)

This section is the main part of the project. In Section 2, I argued that when a climate shock hits, banks protect themselves by charging more and lending less, and that the borrowers who are turned away do not simply disappear. Here, I turn that argument into a model. I first set out the idea and the research question (Section 3.1). Then I describe the simulation phase by phase, along with the reasoning behind each step (Section 3.2). Finally, I report what the simulation produces and discuss how far those numbers can be trusted (Section 3.3). The simulation itself is written in Python in the accompanying Jupyter notebook, and this section explains what it does.

3.1 Idea

The research question

Most traditional papers look at how standard commercial banks price climate risk. My novel angle is to ask what happens *after* a traditional bank prices that risk correctly. Recent empirical evidence from the Indian credit market, by *(Das, A., Majilla, T., & Saurav)*, shows

that after a localised natural disaster such as a severe flood, traditional bank branches systematically raise interest rates to price in the new risk. This is what a prudent bank and a prudent regulator would want to see. However, this credit contraction creates a vacuum. *(Mathur, S., & Sarma, R)* find that unregulated or lightly regulated shadow banks, specifically Non-Banking Financial Companies (NBFCs), rapidly step in and expand their lending in flood-affected areas, absorbing the risky borrowers that the traditional banks rejected.

An NBFC lends money like a bank but is not a bank. It does not accept ordinary demand deposits, and it is regulated less strictly. Lenders of this kind, which sit outside the traditional banking system, are usually called **shadow banks**. Since they are subject to lighter regulation, it is harder for a supervisor to see how much risk has built up inside them. This is the reason why the topic matters for financial stability.

The hypothesis:

The claim that I want to test is the following: **strict climate risk regulation on traditional banks does not eliminate systemic climate risk; it merely forces it into the opaque shadow banking sector.**

To make this claim testable inside a model, I call the firms that lose their bank credit after the flood the *migrating firms*. The hypothesis then has two implications that the simulation can check. The first is that the traditional banks' remaining loan book is safer after the shock than it was before, because the weakest borrowers have been removed from it. The second is that the migrating firms, now paying a risk premium to the shadow banks, default at a far higher rate than the firms that stayed with their banks. If both hold, then the risk measured on the regulated banks' balance sheets has gone down while the risk in the economy as a whole has not.

Why a simulation? And why synthetic data?

There is no public dataset that connects, for every firm, its exposure to floods, its bank's lending decision and its later borrowing from an NBFC. A synthetic economy gets around this. Therefore, the limitation of this simulation is clear. It is a **proof of mechanism**: it shows that the logic of the argument works and what it implies quantitatively under stated assumptions. *It is not a forecast for the Indian economy*. To keep the starting point from being arbitrary, I calibrate the synthetic firms to published Reserve Bank of India (RBI) data on firm financial health. I discuss more on this trade-off in later sections.

Now, I will discuss about each phase of the simulation. The simulation is done in four phases:

- *Phase 1: Generating a synthetic economy*
- *Phase 2: Simulating climate shock and bank reaction*
- *Phase 3: Simulating the Shadow Banking Arbitrage/ Shadow Banking spillover*
- *Phase 4: Visualization and Results*

3.2 Methodology

Instead of observing real firms, I draw their characteristics at random from fixed probability distributions, then simulate a climate shock, and study the resulting outcomes.

3.2.1 Phase 1: Generating a synthetic economy

What we do?

I create 10,000 corporate borrowers, each of which holds a single loan from a traditional bank before the flood. Every firm is described by its revenue, its operating profitability, its debt, and its exposure to floods.

Simulated Firm profile:

Here are the details of the firm profile simulation tabulated in *Table 4*.

Variable	Interpretation
Revenue	The firm's annual sales, in millions of rupees. Drawn from a normal distribution centred on 500 with a standard deviation of 150, and bounded between 100 and 2,000 so that no firm has negative or implausibly large sales.
EBITDA margin and EBITDA	EBITDA is earnings before interest, taxes, depreciation and amortisation. It is the standard measure of the operating profit available to pay lenders. Each firm's margin is drawn uniformly between 10% and 30% of revenue, so firms differ in how efficiently they convert sales into operating profit.
Total debt	The firm's borrowing, expressed as a multiple of revenue. This is the key source of heterogeneity in the model and is explained below.
Flood vulnerability	A score from 1 to 100 describing how physically exposed the firm is to a flood. A score of 100 is the most exposed.
Baseline rate	The 8.5% interest rate that a traditional bank charges every firm in normal times.
Interest expense	Debt multiplied by the interest rate: the annual cost of servicing the loan.
Interest Coverage Ratio (ICR)	EBITDA divided by interest expense. It tells us how many times the firm's operating profit covers its interest bill. An ICR of 4 means EBITDA is four times the interest due; an ICR below 1 means the firm cannot pay its interest out of operating earnings at all.

Table 4: Simulated Firm profile created in Phase 1

Interest Coverage Ratio (ICR) as a central metric:

Section 2 argued that, for a bank, climate risk is ultimately credit risk: the bank is concerned with whether the borrower can keep paying. The ICR is the simplest, widely used summary of exactly that capacity. It is also a natural metric for this study because it responds to both of the forces we care about: a fall in the firm's earnings (the numerator) and a rise in its borrowing cost (the denominator). A flood hits the first, and the response of lenders hits the second.

Calibrating to the RBI evidence

The most important modelling choice in Phase 1 is how debt is assigned. Rather than drawing from one smooth distribution, the firms are split into three groups so that the baseline distribution of Interest Coverage Ratios resembles that of Indian listed non-financial firms in Reserve Bank of India data. In those data, roughly half of firms have a comfortable ICR above 4, a quarter to 30 per cent sit in a middle band between 1 and 4, and a sizeable minority have an ICR of 1 or below. Accordingly, about 50% of synthetic firms are assigned low debt (0.1 to 0.6 times revenue), about 25% medium debt (0.6 to 2.5 times) and about 25% high debt (2.5 to 4.5 times).

Note that the debt is calculated as multiples of revenues. The multiples are chosen like this so that the distribution of Interest Coverage Ratio (ICR) of firms matches the RBI data.

The baseline credit health of the simulated economy is tabulated below:

Baseline credit health	ICR Range	Firms	Share
Strong	ICR above 4	4380	43.8%
Moderate	ICR above 1, upto 4	2899	29.0%
Vulnerable	ICR of 1 or below	2721	27.2%

Table 5: Baseline credit health of the simulated economy (before the flood)

3.2.2 Phase 2: Simulating climate shock and bank reaction

What we do?

A severe flood hits the economy. Firms lose revenue in proportion to their exposure, and traditional banks respond in two ways: they charge higher rates to exposed firms, and they refuse to renew the loans of firms whose repayment capacity has fallen below a safety threshold.

Step 1: The physical shock to firm income

Each firm's revenue falls by a share equal to its vulnerability score scaled to a maximum of 35%. A firm with a score of 100 loses 35% of its sales, a firm with a score of 50 loses 17.5%, and a firm with a score of 2 loses almost nothing. EBITDA falls with revenue because each firm keeps its original margin. This is the channel described in Section 2.2 under physical risk: a climate event damages the borrower's income, and through income its ability to pay. Across the economy, aggregate operating profit falls by about 18%.

Two simplifications are worth stating:

First, the shock is a single, one-off, linear loss of income; it does not model damage to collateral, repair costs or recovery over time.

Second, because margins are held fixed, costs fall one-for-one with sales. In reality fixed costs would make profits fall by more than revenue, so the simulated shock is, if anything, a conservative one.

Step 2: Banks reprice risk

Banks respond to the new risk by raising the interest rate by 50 basis points (0.5 percentage points), from 8.5% to 9.0%, for firms with a vulnerability score above 50. This mirrors the empirical finding that traditional bank branches raise rates after a localised disaster. The logic is that exposed borrowers are now more likely to default, so the bank charges a higher price to compensate. The repricing is intentionally modest: the simulation is not built on an extreme banking response.

Step 3: Banks ration credit

Pricing alone is not how a regulated bank manages risk. The second response is to decline to renew the loan of any firm whose post-shock ICR, calculated using the new earnings and the new interest rate, falls below 1.5. The threshold reflects a common benchmark: a coverage ratio below 1.5 is widely read as a sign of acute financial stress, because the firm has little cushion between its operating profit and its interest bill. We interpret the cut-off as the combined effect of prudential supervision and climate-adjusted credit appraisal, in line with the regulatory pressure on banks discussed in Section 2.7.

The decision rule is deliberately *correct* from the bank's point of view. Banks reject exactly those firms that have become too risky to lend to. This matters for the central argument: we are not assuming that banks are careless or irrational. We are asking what happens to the economy when banks do exactly what regulators want them to do.

In this run, 5,896 firms pass the test and keep their bank loan, and 4,104 firms (41%) are rejected. Of the rejected firms, about 90% already had a baseline ICR below 1.5 before the flood, so the flood tips mostly borderline or already weak firms over the edge, rather than turning healthy ones into rejects.

3.2.3 Phase 3: The Shadow Banking Spillover

What we do?

Firms rejected by traditional banks still need finance to continue operating, so they turn to the shadow banking sector, in practice Non-Banking Financial Companies (NBFCs). I assume that NBFCs lend to all rejected firms but charge a climate risk premium of 400 basis points over the baseline rate, so rejected firms now pay 12.5%, up from 8.5%. Firms that stay with their banks continue to pay the bank's post-shock rate.

The logic of the assumptions

- **Full absorption.** The model assumes that NBFCs absorb every rejected firm. This follows the evidence, that NBFCs expand lending sharply in flood-affected areas, and it isolates the mechanism of interest: the question is not whether the risk is lent against but at what price and with what consequences.
- **A risk premium.** A lender willing to take on borrowers whom banks have judged too risky will demand compensation. The 400 basis point premium is a plausible but illustrative magnitude, and it is the most influential parameter in Phase 3. We discuss its role in Section 3.3.
- **Weaker oversight.** Because NBFCs are lightly regulated, the build-up of risk on their books is harder for a supervisor to observe. In the simulation, this appears as the fact that the defaults occur on shadow bank loans, a portfolio that does not appear in the bank-level statistics regulators monitor.

Defining default

A firm is classified as defaulting if its final ICR, computed using the post-shock earnings and the interest rate it now pays, falls below 1.0. At that point, operating profits are insufficient to pay even the interest on its debt, and without new capital the firm cannot continue to service the loan. This is a technical, or interest-coverage, definition of default. It captures the inability to pay out of operating income, and does not model bankruptcy proceedings, restructuring, asset sales, or the possibility of fresh equity injections. We therefore read the output as a measure of financial distress, rather than a literal count of legal failures.

3.2.4 Phase 4: Visualisation and Results

The final phase moves from individual firms to economy-wide totals to show the macroeconomic impact. Three visualisations summarise the results. The first is a bar chart of loan portfolios before and after the flood, showing the movement of loans from banks to NBFCs. The second is a pie chart of the survival and default rates within the shadow banking portfolio. The third is a density plot comparing the distribution of interest burdens before and after the shock. Each corresponds to one of the claims in the hypothesis: that credit migrates, that migrated credit is fragile, and that the burden of debt servicing rises across the economy.

Plot 1:

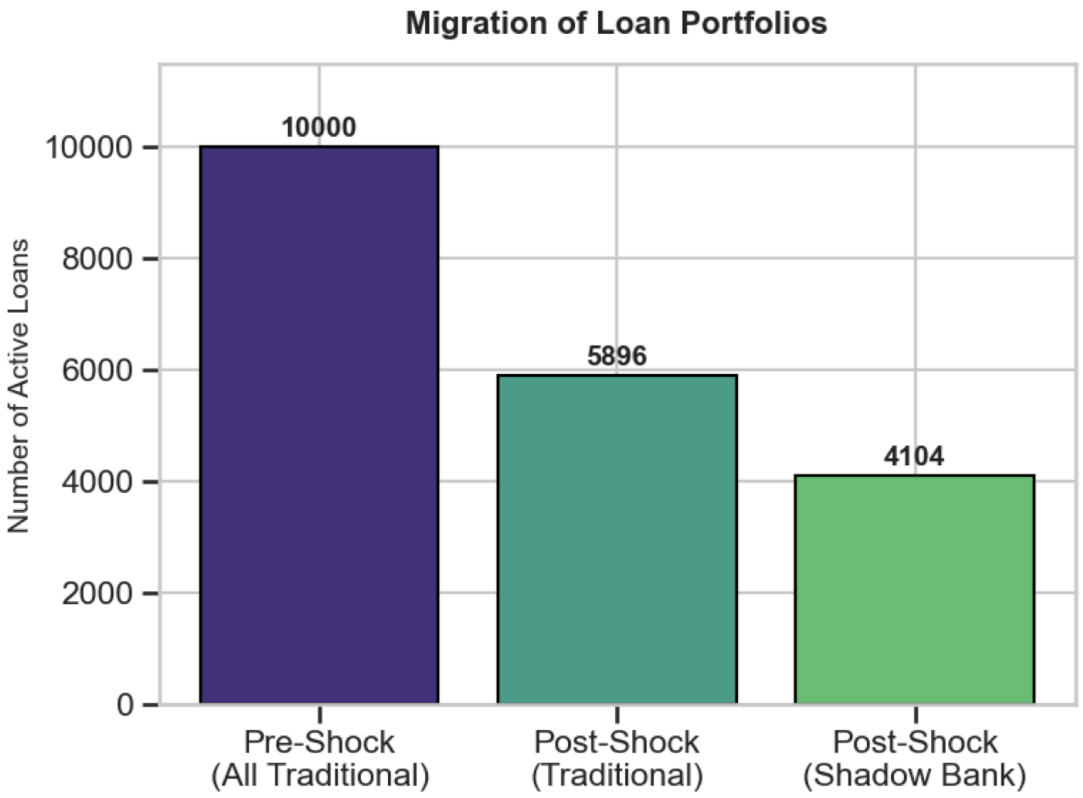


Figure 3: Plot 1

Plot 2:

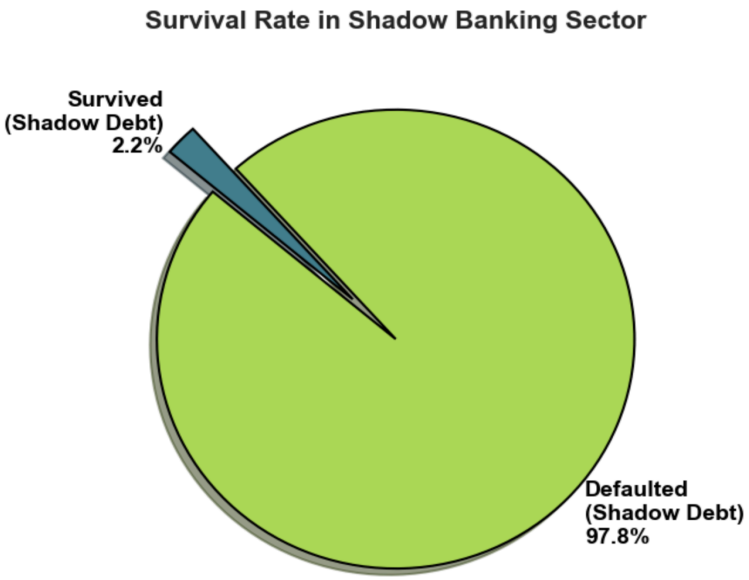


Figure 4: Plot 2

Plot 3:

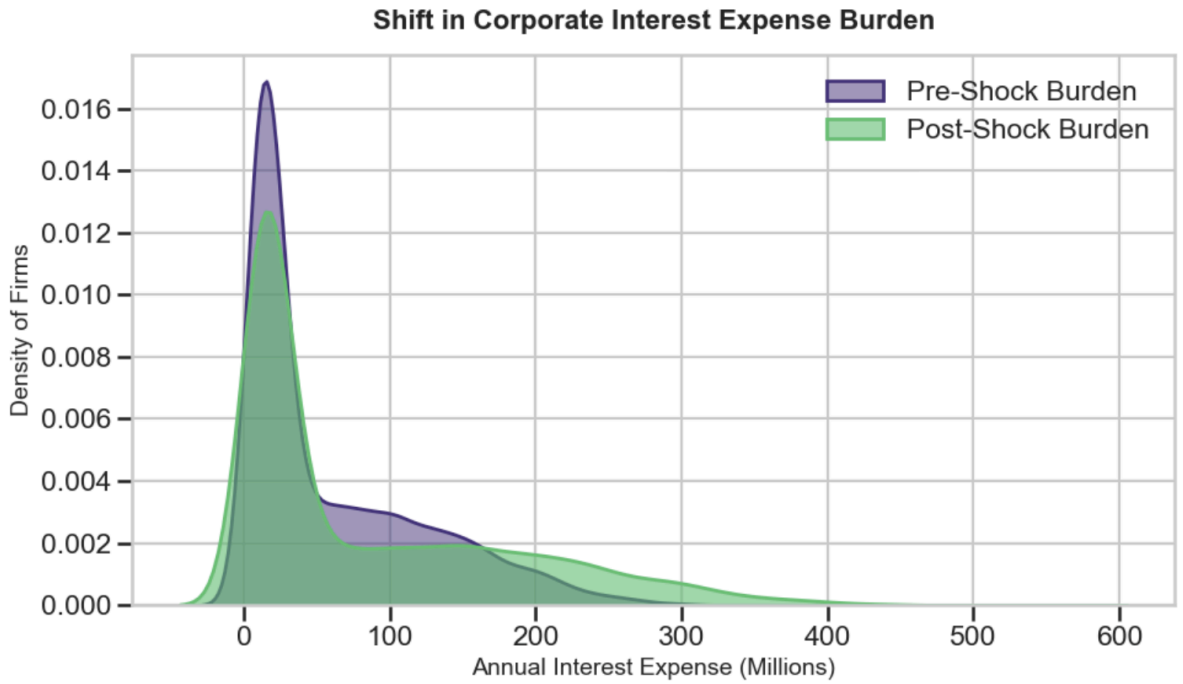


Figure 5: Plot 3

Results:

- I simulated a synthetic economy of 10000 firms to test regulatory climate risk behavior.
- After a simulated severe flood shock, traditional banks aggressively de-risked. They hiked interest rates by 50 basis points for highly vulnerable firms and outright rejected 4104 firms whose Interest Coverage Ratios dropped below 1.5.
- While the traditional banking sector looked safer on paper, the macroeconomic risk did not disappear. It migrated.
- Unregulated Shadow Banks (NBFCs) stepped in to absorb all 4104 rejected firms, but charged an exorbitant 400 basis point risk premium.
- Because highly vulnerable firms were forced to take on extremely expensive shadow debt just to survive the shock, 4013 of them collapsed into default.
- *Conclusion: Traditional bank climate stress tests are fundamentally flawed if they ignore shadow banking. Pushing climate risk out of regulated banks actually increases total systemic defaults.*

3.3 Discussion

Credit migrates. Before the flood all 10,000 loans sit with traditional banks. Afterwards, 5,896 remain with banks and 4,104 move to NBFCs *(Figure 3)*. Because the firms that move

are on average the most heavily indebted ones, the shift in volume is far greater than the shift in the number of loans.

The bank book looks safer. The firms that remain with banks have a higher average pre-shock ICR, compared with the original economy as a whole, and the share of bank borrowers with a strong ICR (above 4) rises, this is due to the fact that, post the climate shock banks refuses to provide credit to firms below the cutoff ICR. This is the first implication of the hypothesis: a stress test looking only at the regulated bank would see a stronger loan portfolio after the flood than before it.

The migrated risk is fragile. Of the 4,104 firms that move to NBFCs, 4,013 (97.8%) default and only 91 survive *(Figure 4)*. Those firms had ICR below 1.5, so most were weak before the flood. Defaults are therefore concentrated entirely in the shadow banking sector. This is the second implication of the hypothesis.

The cost of debt rises economy-wide. Total interest payments across the economy rise *(Figure 5)*. The rise is driven by two things: the higher bank rate for exposed firms and, above all, the move of heavily indebted firms to a lender charging high credit premiums. The density plot shows the same thing in distributional terms, with the right tail of large interest burdens becoming thicker.

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